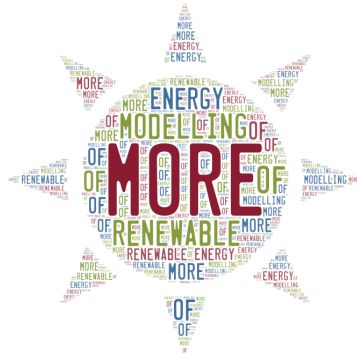




MORE – Modelling of Renewable Energy Software Documentation

Technical Report for:

Projekt 4N: Nordwest Niedersachsen Nachhaltig Neu
TV-7 Landwirtschaftlicher Strukturwandel und Klimawandel-Anpassung

Version of July 24, 2026

Dr. Uwe Reimer / uwe.reimer@hs-emden-leer.de

Hochschule Emden/Leer · Constantiaplatz 4 · 26723 Emden ·
<http://www.hs-emden-leer.de>



Contents

1	Introduction	1
1.1	What is MORE?	1
1.2	Required software	2
1.3	How to run a simulation	2
1.4	The results page	4
2	Software documentation	6
2.1	Program files	6
2.2	Internal data structure	6
2.3	Internal workflow	8
3	General model description and example use case	10
3.1	Technical components	11
3.2	Economic and Ecological model	15
3.3	Example use case	18
4	Appendix	25
4.1	Calculation of the energy demand for the compression of hydrogen	25
4.2	German energy efficiency ranking for residential buildings	26
4.3	Literature survey for CAPEX	27
	References	28

1 Introduction

1.1 What is MORE?

MORE is a system simulation tool that uses data for local power generation from photovoltaics and wind from an external website (PVGIS, photovoltaic geographical information system) [1]. Based on this information, MORE lets you design and operate an energy system with the following features, which are visualized in Figure 1.

- Local renewable energy production
- Distribution to hydrogen production and storage or grid feed-in
- Using hydrogen for mobile applications based on a weekly driving cycle
- Using hydrogen for heat demand (application for buildings)

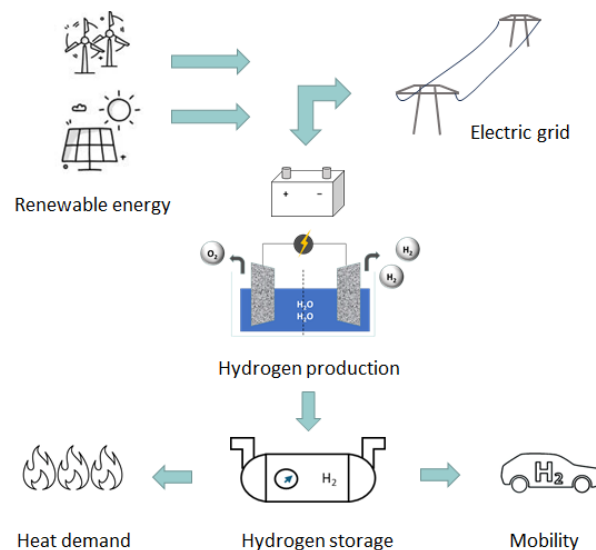


Figure 1: Overview for the system model.

MORE is written in PYTHON, data export occurs as HTML page or *.csv files, data import from *.csv files and text files. The software was developed under MS Windows, but it should run on any computer with PYTHON installed.

1.2 Required software

The following software needs to be installed on your computer.

- Python [2] (version 3.12.4 was used for development)
- Matplotlib for Python [3]

For the data from PVGIS [1], choose for your location:

- HOURLY DATA
- PV Power
- Installed Peak PV power = 1 kWp

1.3 How to run a simulation

First, download the software package [4] on your computer. After unpacking, the main directory contains the following files and directories.

/moresystem This directory contains the main simulation files and additional input files.

01_input.txt Input, which is defining the system.

02_RunSystem.bat Executable script (MS Windows) that starts the simulation.

03_results.html This file is generated after successful simulation and contains the results.

04_errors.txt Here is the output for Python error messages. Normally, this file should be empty.

***.png** (13 files) These files are generated after successful simulation and contains results.

The MORE software simulates an energy system. The input for the system simulation is defined in three separate files. Figure 2 shows a sketch of the overall structure.

01_input.txt Input, which is defining the system.

/moresystem/Emden-2023.csv The filename is defined in '01_input.txt'. It contains Data from the PVGIS website: PV power, wind speed, temperature. (Time resolution needs to be one hour!).

/moresystem/fuel-week.txt The filename is defined in '01_input.txt'. It contains the fueling plan for mobile hydrogen usage (Time resolution is one day, overall time is one week).

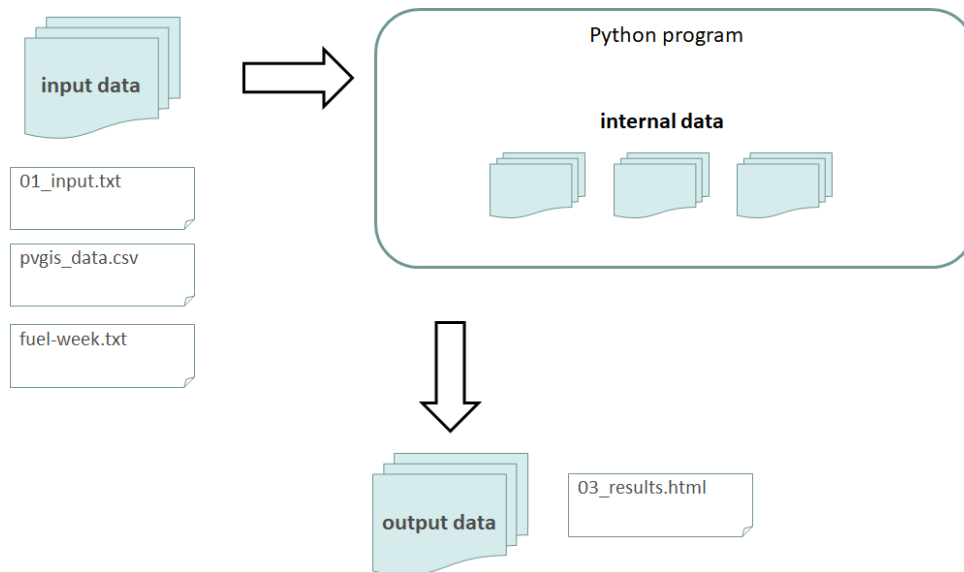


Figure 2: Overall flow of information.

In order to run a simulation, the following steps need to be performed.

1. Check the content of the three input files and modify them in order to create the desired system.
2. Double click on the file '02_RunSystem.bat' or execute the script. You may use the command from this file to run the program from the command line.
3. After completion, the HTML file for results is updated. In the unlikely event of an error occurring, the HTML file is not updated and an error message can be found in the file '04_errors.txt'.
4. (Optional) You can export all data which is shown on graphs into *.csv files (see bottom of file '01_input.txt'). Use these files for further analysis or create your own plots.

1.4 The results page

The file '03_results.html' contains all results of the simulation. If it is open in a browser, make sure to refresh the page after a new simulation run. The page is separated into several sections. Figure 3 (Left) shows the upper part with the major results. In Figure 3 (Right) the lower part with additional information and internal state of the system is displayed.

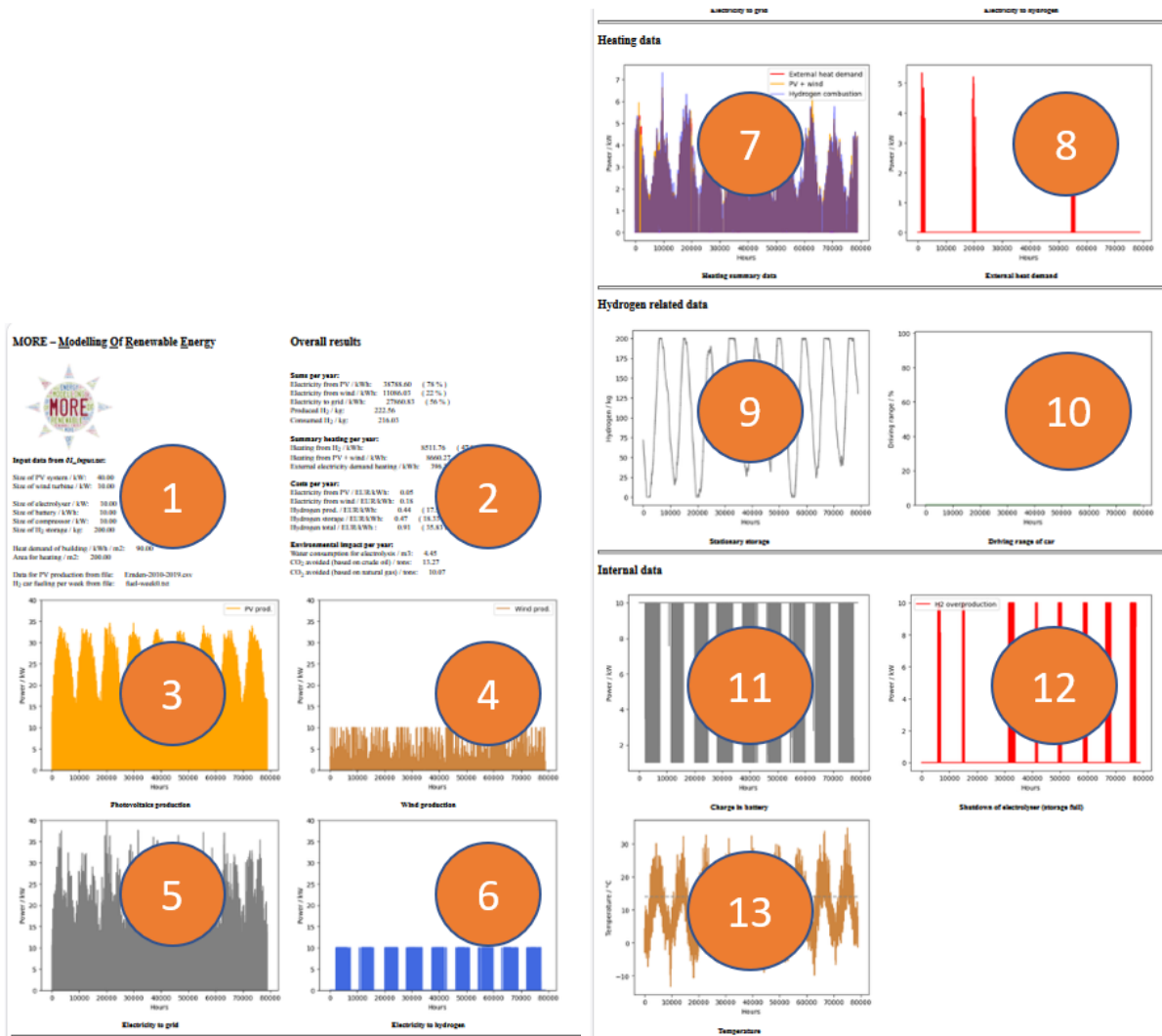


Figure 3: The results page – Left: major information; Right additional information.

The following information is summarized on the HTML page (see numbers in circles).

1. Summary of input data for the present simulation
2. Average values per year for production, consumption, costs and environmental impact
3. Photovoltaics production
4. Wind power production
5. Electricity exported to grid
6. Electricity converted to hydrogen (operation of electrolyser)
7. Heating data with contribution from hydrogen combustion, PV+wind and external demand
8. External demand for heating (This is useful to see at which times the system fails to be self-sufficient.)
9. Level of the hydrogen storage
10. Achieved driving range for mobile application
11. Level of battery storage (The battery acts as a buffer for the electrolyzer.)
12. Shutdown events of the electrolyzer (storage is full)
13. Outside temperature (as imported) with dotted line for the onset of heating

(Optional) All data for the graphs can be exported to *.csv files. The file names will be identical to the graphs. (This is defined in the file '01_input.txt'.)

2 Software documentation

2.1 Program files

The program files and source code are documented in the file '00_Readme.Files.html' within the folder '/docu'. The overall structure is shown in Figure 4.

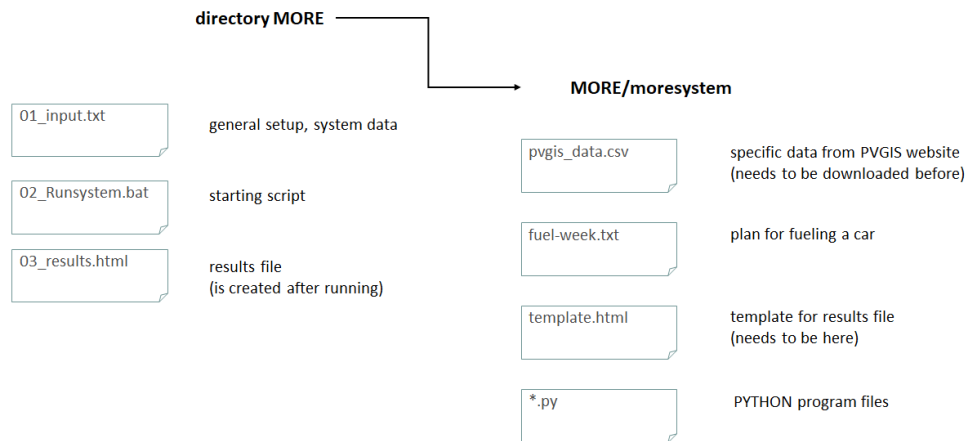


Figure 4: Overall file structure.

2.2 Internal data structure

Input file

The input file is arranged in sections. The name of the section is written like this: `### system ###`. Each section contains parameters in the following format: `name = value (digit) = description`. Here is an example.

```
### system ###
```

```
...
```

```
sizeBA = 50.0 = size of battery in kWh
```

```
sizeCO = 50.0 = size of compressor in kW
```

```
sizeTA = 100.0 = size of hydrogen tank in kg (H2)
```

```
offset = 4.0 = starting day for the year / 2010 starts with Friday = 4 (otherwise set this to 0)
```

```
### operation ###
```

```
elim = 0.2 = % / limit -> only operate electrolyser above this value (20% of max load is recommended)
```

```
batlim = 0.1 = % / limit for discharge of battery (0.1 = 10%)
```

```
se = 60.0 = kWh / kg H2 -> hydrogen production / efficiency of electrolyser + components = 70%
```

```
hhv = 39.4 = kWh / kg H2 -> hydrogen thermal usage / higher heating value
```

```
### wind ###
```

```
height = 15.0 = hub height in m
```

...

The content of this file is read into a dictionary, where the key is the name of the section, e.g. 'system', and the values are imported as an ordered list. Here, the names of the parameters are ignored in the internal data structure. For your own work you can extend this file by adding any number of sections with any number of parameters. The software MORE will provide this data through the dictionary **myIn**. You just have to add some code to do something with this additional parameters ...

Results data

All results are collected in the list **myRes**. Here is an overview over the available data at the end of the simulation run.

- 0 power for H2 generation / kWh
- 1 electricity to grid / kWh
- 2 charge in battery / kWh
- 3 heating demand from hydro / kWh (this is full demand minus available PV+wind)
- 4 produced H2 / kg (based on efficiency of electrolyser)
- 5 consumed H2 / kg (heating) /to this is the same as [3] just in kg (based on higher heating value)
- 6 tank H2 in stationary / kg
- 7 H2 in car / kg
- 8 H2 overproduction (cannot be stored) / kg
- 9 additional heat demand (storage empty) / kWh
- 10 Driving range of car / percent
- 11 Heating by stored hydrogen / kW
- 12 Heating by PV + wind / kW

Usually, each entry contains a simple list of the specific result. The length is equal to the total number of hours in the simulation run.

Other data

All other data, which is needed for internal calculation, is stored in these lists.

dataPV PV power / kW (recalculated for the actual system size defined in input)

dataWind wind power / kW (for system size from input)

dataTemp temperature (data from PVGIS website)

dataHeat heat demand of building

dataCar fueling demand of car (kg hydrogen)

2.3 Internal workflow

The main routine is defined in the file `/moresystem/myrun.py`, which is called by the starting script.

1. Initialization: At the beginning all other modules are loaded, files and the working directory are defined.
2. Reading input: Input data is read from files. Data for heat demand and/ or hydrogen consumption for mobile are prepared in order to fit the timeframe of power production.
3. Initial power distribution: Power production is calculated from wind and photovoltaics. The power is distributed to:
 - power for H2 generation
 - electricity to grid
 - charge in battery
 - direct heating

The last point follows the logic that if there is heat demand, electricity will not be converted to hydrogen, but used directly for heating. This avoids the conversion losses of the electrolyser. Electric heating is assumed through an ohmic resistor, i.e. no heat pump system.

4. Calculate the mass of hydrogen: Here, the mass of produced hydrogen and the mass for heating demand are calculated. (Mass of produced hydrogen is needed for storage. Mass of hydrogen demand for mobile is already given in kg).
5. Hydrogen distribution: Hydrogen is distributed to storage and for mobile consumption (car fueling). If storage is full, the amount of 'overproduction' is calculated. This is a theoretical value, since the electrolyser would not be operated if hydrogen cannot be stored. Hence, value of produced hydrogen needs to be corrected in the following step. In the case of heating: if the heat demand cannot be met by the available hydrogen, then there is an additional power demand (external heating demand).

-
6. Correction for limited storage: If storage is full, no hydrogen is produced. The amount of 'overproduction' is shifted to the grid.
 7. Driving range: Here, the driving range for mobile application is calculated in percent. The value of 100% means, that the defined drive cycle (i.e. daily fueling amount) is achieved.
 8. Analyze heating: The software considers three contributions to heating, namely hydrogen combustion, wind+PV and external heat demand.
 9. Plotting of results: Generate all required graphs. In the case of heating it might be useful to exclude the first year of the simulation in the reporting, since the year starts in January with an empty hydrogen storage. (This is defined in the file '01_input.txt'.) Actually, any number of years can be excluded.
 10. Calculate sums per year: Calculate yearly sums for reporting on the HTML page.
 11. Generate results page: Generate the HTML page for results.
 12. Export of results data: (Optional) All data for the graphs is exported in *.csv files. The file names are identical to the graphs. (This is defined in the file '01_input.txt'.)

For more details of specific functions and modules, check the file '00_Readme_Files.html' within the folder '/docu'.

3 General model description and example use case

The following part of this document is taken from the final report of the project, which was published in German language. The translation was guided by the tool 'ChatAI Qwen 3'.

Within the scope of this work, a decentralized energy system for the production and utilization of green hydrogen is investigated. The simulations were carried out using the software MORE (Modelling Of Renewable Energy) developed within the project [4]. Figure 5 shows the general flowchart of the simulation and the components considered. Electrical energy is generated in a decentralized manner using photovoltaic systems or wind power. This energy is available for the production of green hydrogen, which is realized via an electrolysis unit (supported by a battery as a short-term storage system). It is economically impractical to size the electrolysis unit based on the maximum electricity output of the renewable energy converters. Therefore, any excess electrical energy exceeding the electrolysis capacity is fed into the general power grid. The produced hydrogen can be used locally if an additional storage tank is available. The size of the storage tank is determined economically based on the intended application, as will be demonstrated in the subsequent sections. Hence, Figure 5 illustrates only the simplest conceivable scenario, in which the generated hydrogen is injected into a theoretically available gas network. This approach is equivalent to assuming an infinitely large intermediate storage (natural gas network) that is available at no cost.

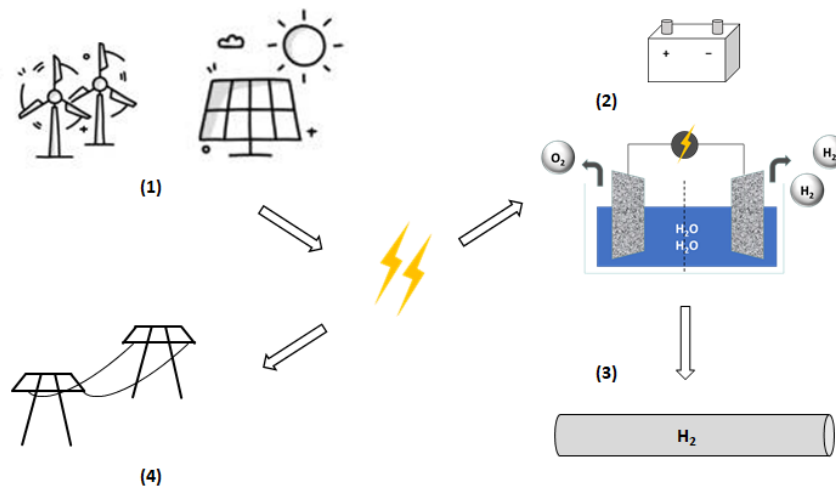


Figure 5: Sketch of the system simulation with these components: (1) photovoltaics and wind power, (2) elektrolyzer with battery, (3) hypothetical gas pipeline (symbolic for storage or consumption), (4) electric grid.

3.1 Technical components

Photovoltaics

The European Commission provides a web platform that enables access to meteorological data as well as the performance calculation of photovoltaic (PV) systems based on their geographical location [1]. Input parameters required include the geographical location, the size of the PV system in kW, the tilt angle, the orientation angle (azimuth), and the desired time period (one or multiple years). The PVGIS web platform [1] generates, based on satellite data, a file with hourly resolution containing the following information, with only the data relevant for the simulation being listed here.

- PV power
- Temperature
- Wind speed at 10 m

For the simulation, a reference PV system size of 1 kW is assumed, which is then scaled within the system model according to the system's specific requirements. The PV modules are oriented towards the south with a tilt angle of 23°. The resulting simulation data were compared with measured data from a 40 kW PV system installed at the University of Applied Sciences Emden-Leer. The deviations for the reference year 2020 are less than 10%, which represents sufficient accuracy for the purposes of this study.

Wind turbine

The power output of the wind turbine is calculated based on the wind speed at a height of 10 m. This approach provides only a rough estimate, as the exact power curve of any specific wind turbine is not known to the system model. Furthermore, the power output depends not only on wind speed but also on wind direction (e.g., wake effects from buildings, trees, etc.) and the installation height. The parameters used in this study were obtained from measurements conducted on an Enercon E-18 wind turbine, which is operational at the University of Applied Sciences Emden-Leer [5]. While these parameters are not generally applicable to other locations, they yield reasonable and useful results for estimating wind energy yield within the scope of the system model. The model requires the following input data: turbine hub height, nominal power output, rated wind speed, cut-in wind speed, and cut-out wind speed.

In order to calculate the power output, the wind speed at the turbine's hub height is first determined. This is calculated using the wind profile equation 1, also known as the Hellman equation [6, 7, 8].

$$v(h_2) = v(h_1) * \left(\frac{h_2}{h_1} \right)^\alpha \quad (1)$$

The value $\alpha = 0.13$ is a measure of the so-called surface roughness at the location of the University of Applied Sciences Emden-Leer. The power output is determined using the power curve (Equation 2), which corresponds to the Enercon E-18 turbine [5]. The use of alternative power curves is possible within the simulation program at the level of the source code.

$$P = 0.09 * v(h_2)^{2.82} \quad (2)$$

For a more accurate calculation, local measurements at the specific site are necessary, which would additionally account for directional dependencies. However, this goes beyond the scope of the current system model. The simulation results were validated against measured data from a real wind turbine (Enercon E-18 at the University of Applied Sciences Emden-Leer). For the reference year 2020, a deviation of less than 5% was observed for the annual total energy yield, although daily deviations were significantly higher. For the purpose of estimating energy system performance, this level of accuracy is considered sufficient. Since the Enercon E-18 is no longer available on the market, for illustrative purposes the parameters of a 20 kW turbine model 'CF20' [9] with a hub height of 15 m are used in the simulation, as this height is exempted from the legal approval process in northern Germany [10].

Battery storage

The battery storage is considered an ideal storage device within the model. As indicated in Figure 5, it serves to support the electrolysis process. On one hand, it compensates for fluctuations arising from electricity generation by the photovoltaic (PV) or wind power plant. In the case of the PV system, the battery also enables a 'prolonged' utilization of the daily PV power output. The battery is operated at all times above a certain threshold, which is set at 10% of its nominal capacity in this study.

Electrolyzer

The production of 'green' hydrogen is achieved through the electrolysis of water (see Equation 3), with the electricity supplied from renewable energy sources.



Currently, there are two main different technologies for water electrolysis. Alkaline electrolysis has been used in industry for many years. For systems in the megawatt (MW) range, the efficiency typically ranges from 52% to 71% [11, 12]. In contrast, PEM electrolysis (Proton Exchange Membrane, also known as Polymer Electrolyte Membrane) has only been available on the market for a shorter period. It is characterized by a simpler system design and the ability to operate very dynamically [13]. Currently, efficiencies of 44% to 67% can be achieved with PEM electrolysis [11]. However, unlike alkaline electrolysis, PEM electrolysis requires noble metals as catalyst materials, which currently offset the cost advantages offered by its simpler system architecture [11].

Currently, only very limited real-world data are available for small-scale systems. During the multi-year operation of a small H₂ refueling station, an energy demand of approximately 60 kWh/kg H₂ was determined, corresponding to an efficiency (based on the lower heating value, LHV = 33 kWh/kg H₂) of about 55%. For larger systems, energy demands of around 50–55 kWh/kg H₂ are typically achieved [14, 12]. For the calculations in this study, an energy demand of 60 kWh/kg H₂ is used,

which already includes the energy required for compression. As an additional technical constraint, the electrolysis unit is not operated below 20% of its nominal load.

Hydrogen storage

Hydrogen storage can, in principle, be realized in three ways: in liquid tanks, in high-pressure tanks, and in the form of bound metal hydrides [15]. For the scenarios considered in this work, the storage system should be commercially available and applicable to smaller-scale installations. Currently, high-pressure tank storage is the only option that meets both requirements and is therefore used in this study.

Hydrogen storage tanks are primarily available in two pressure levels: 200 bar or 350 bar. A higher storage pressure, such as 700 bar used in the automotive sector, is not suitable in this context, as hydrogen no longer behaves like an ideal gas at such high pressures, leading to a significant increase in the required compression energy. To calculate the compression energy, an overpressure of 25 the tank [16]. Generally, the compression energy can be estimated at approximately 10% of the higher heating value (HHV) of hydrogen (about 4 kWh kg⁻¹) [17]. For a more accurate assessment, the compression work can be calculated using thermodynamic principles (see Appendix), which confirms the value of 4 kWh kg⁻¹.

In technical applications, the size of the compressor should be approximately matched to that of the electrolysis unit, as the produced hydrogen gas must be compressed to the corresponding storage pressure (or pressure required for injection into a pipeline network). In this model, the compressor's energy demand is not calculated separately but is instead included as part of the electrolysis process. The storage tank is modeled here as a conventional steel pressure vessel capable of operating at 200 bar.

Consumption – Electromobility

In the simplest case, electricity from renewable energy sources can be directly used in electric vehicles equipped with batteries. The main advantage is that nearly no losses occur during this process. However, the disadvantages include relatively long battery charging times and the additional weight of the vehicles, which becomes increasingly disadvantageous for higher-performance vehicle classes. In agricultural applications, it is often crucial that vehicles are available throughout the entire day. Direct charging from a PV system is therefore impractical, and in such cases, an additional battery would need to be used as an intermediate buffer. The use of fuel cells in vehicles would solve this issue. This would require an electrolysis unit – typically as expensive as an additional battery – and the produced hydrogen can be stored and remains available for use even days or months later. Table 1 lists the hydrogen consumption of various vehicles to enable an estimation of a hypothetical demand in agricultural operation. Currently, fuel cell vehicles are not yet available on the market for agricultural use. Therefore, the hydrogen consumption values in Table 1 were calculated based on the following relationship: a small tractor (engine power output of approximately 35 kW or 50 hp) consumes about 80 liters of diesel per day, while a large tractor (engine power output of approximately 220 kW or 300 hp) consumes about 400 liters of diesel per day [18]. Unlike conventional road vehicles, where

fuel consumption is typically expressed per 100 km, agricultural machinery consumption is based on a full working day rather than distance traveled. Consumption values vary significantly depending on the specific tasks performed. Hence, Table 1 provides average values to represent typical daily requirements.

Table 1: Hydrogen demand of selected fuel cell vehicles.

Vehicle	H ₂ consumption	Source
Passenger car	approx. 1 kg per 100 km	[19]
City bus	approx. 10 kg per 100 km	[20, 21]
Tractor small	approx. 20 kg per day	[18]
Tractor big	approx. 100 kg per day	[18]

Based on the values from Table 1, it is possible to estimate how many times per week refueling would be required. The system model employs a weekly profile in which a specific amount of hydrogen is withdrawn from the storage tank for each day of the week.

Consumption – Heating

The energy efficiency class of a building [22] indicates how much energy is required annually for heating and hot water preparation. The average annual heat demand in Germany has remained relatively constant since 2014 at approximately 150 kWh/m² [23], which corresponds to energy efficiency class 'E' for buildings (see Table 11 in the appendix).

The model calculates the hourly heat demand based on this annual total, assuming that the heat demand is primarily caused by residential heating. The heat loss of a building is proportional to the temperature difference between the indoor and outdoor temperatures. Both values can be specified within the simulation. The model uses the local weather data from the PVGIS web platform [1] (see the photovoltaics) and automatically determines the appropriate thermal resistance (insulation) of the building so that the required annual total heat demand is achieved.

The heat demand is met by the model through combustion of hydrogen, which can be implemented by retrofitting conventional heating systems (gas or oil). Theoretically, a more efficient approach would be to regenerate electricity using a fuel cell and subsequently utilize the energy via a heat pump. Although fuel cells have an efficiency of only about 1/3 (plus usable waste heat), modern heat pumps can achieve seasonal performance factors of 3 to 4, meaning three to four times more heat is obtained compared to the electrical energy input. However, retrofitting older buildings for such systems is often difficult and associated with significant additional costs. Including both a fuel cell and a heat pump in the model would considerably increase its complexity. When comparing the energy output of both heating variants, a rough estimate suggests that in both cases a similar amount of stored hydrogen is required. At least for a general assessment, thermal utilization is sufficient. Therefore, the model calculates thermal utilization based on the higher heating value of hydrogen, which is 39.4 kWh/kg. Figure 6 (left) shows the temperature profile for the location of Emden in 2023. The dashed line represents the threshold (here $T = 14^{\circ}\text{C}$) above which heating should be

activated. The indoor temperature is fixed at $T = 20^\circ\text{C}$ in this case. The right-hand image shows the calculated heat demand for a building with an annual demand of 90 kWh/m^2 (energy efficiency class 'C') and 200 m^2 of living space.

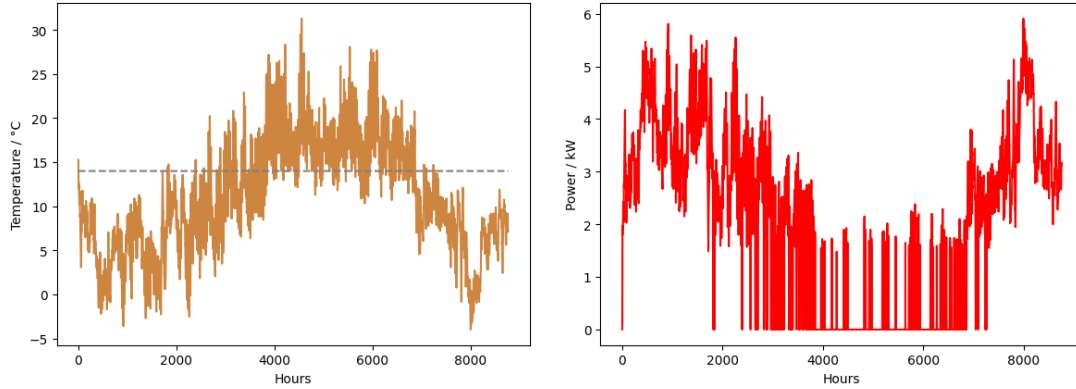


Figure 6: Left: temperatur at the city of Emden in 2023; Right: resulting heat demand for a building with 200 m^2 living space (energy efficiency class 'C').

3.2 Economic and Ecological model

Economic analysis

For estimating costs, it is necessary to account for the contributions of individual system components. The investment costs (CAPEX) for the model are summarized in Table 2. Although the literature data in Table 12 (see Appendix) shows considerable variation, reasonable values can still be derived for the target size in the two-digit kW range, enabling a meaningful estimation of total costs.

Photovoltaic systems, wind turbines, and batteries have been established on the market for some time, and their prices can be compared with current offers available on the websites of several companies. When considering wind turbines, it is necessary to distinguish between two power ranges. In large wind parks, turbines with capacities exceeding 1 MW are now commonly installed. This power class has already undergone significant optimization due to high production volumes and is therefore quite cost-effective. However, such large-scale turbines are not suitable for decentralized deployment on smaller plots of land. Smaller turbines (below 500 kW) are available in small production series but are significantly more expensive. Electrolyzers are currently predominantly used in large-scale industrial applications. For the present study, prices for systems in the kW range are estimated based on available literature data. Electrolyzers in the two-digit kW range have so far only been produced in small series or as prototypes. The same applies to hydrogen compressors.

Since operational costs (OPEX) are typically much lower than investment costs, they are estimated in this study as a flat 2% of the annual costs [24, 25]. The annual costs are calculated here from the investment costs (Table 2) divided by the depreciation period. It becomes evident that the length of the depreciation period is a crucial parameter influencing the final costs. In this study, the following values are used: 25 years for photovoltaic systems and wind turbines [26], and 10 years for

Table 2: CAPEX used in the model.

Component	Cost	Capital allowance
Photovoltaics	1300 EUR/kW	25 years
Wind turbine	5000 EUR/kW	25 years
Elektrolyzer	1500 EUR/kW	10 years
Compressor	1500 EUR/kW	10 years
Battery	800 EUR/kWh	10 years
Tank	approx. 13 EUR/kWh (500 EUR/kg H ₂)	25 years

batteries [26]. A depreciation period of 25 years is also applied to the hydrogen storage tank, while electrolyzers and compressors are assumed to have a 10-year lifespan.

From Table 2, it is evident that storing energy in the form of hydrogen is significantly more cost-effective than storing it in a battery. Nevertheless, storage remains a major cost factor, which can be illustrated using the previously mentioned example of a house's heating demand. As described in the preceding section, a house with 200 m² of living space and an annual energy demand of 90 kWh/m² has a heating demand of 18 000 kWh. If this entire amount were to be stored, for instance as a replacement for an oil heating system, the required investment for a high-pressure hydrogen storage tank would amount to 234 000 EUR. There is considerable potential for cost reduction—especially, since much cheaper high-pressure tanks are already being developed for the automotive sector. There, costs as low as 3 EUR/kWh (120 EUR/kg [27]) may be achievable. Nevertheless, the cost of storing hydrogen in a high-pressure tank is still currently about 130 times higher than storing oil, which costs approximately 0.10 EUR/kWh. Therefore, it appears more sensible not to store the entire annual hydrogen production as a reserve. Possible alternatives include using a fraction of the hydrogen for fuel cell vehicles or selling it seasonally.

Ecological analysis

CO₂ Reduction and CO₂ Avoidance Costs

Both terms – reduction and avoidance costs – can be used to quantify the impact of green hydrogen usage on climate balance in numerical terms. In this study, the following distinction between the two terms is applied:

CO₂ Reduction refers to the emissions from fossil energy carriers that are replaced by the use of green hydrogen.

CO₂ Avoidance Costs are defined as the difference between the costs of using green hydrogen and the costs of the fossil fuel it replaces, divided by the amount of CO₂ savings. In simpler terms, this represents the ratio of the additional costs of green hydrogen to the CO₂ reduction achieved (in EUR per metric ton of CO₂, or EUR/t_{CO2}).

In this study, only CO₂ reduction is determined, as the costs of the fossil technologies being replaced are not within the scope of the investigation and can vary significantly depending on the specific application. Furthermore, CO₂ reduction is not independent of the type of fossil fuel involved, as illustrated in Table 3. For the application cases discussed subsequently, crude oil based fuels and natural gas are relevant. Crude oil, heating oil, diesel, and gasoline all have similar CO₂ emission levels. Therefore, for the evaluation of the results, a reference to crude oil and natural gas is adopted.

Table 3: CO₂ Emissions of fossil energy sources [28].

Energy source	Emission in kg CO ₂ /kWh
Hard coal	0.346
Lignite	0.356
Crude oil	0.263
Heating oil	0.288
Diesel	0.266
Gasoline	0.263
Natural gas	0.202

Water demand

The production of green hydrogen involves splitting water into oxygen and hydrogen. Theoretically, approximately 9 liters of pure water are required per kilogram of hydrogen produced [12]. Currently, water for electrolysis is mostly used in drinking water quality [29]. However, at large industrial scale, water consumption is significantly higher, as water is also used for cooling purposes. Depending on the process configuration, about 30 liters [29] or between 20 to 40 liters of water per kilogram of hydrogen may be required [30]. Using drinking water for large-scale hydrogen production can be problematic if local water resources are limited [29]. Moreover, it is important to distinguish between water withdrawal from the environment (total volume used for hydrogen production) and the actual water consumption by the electrolysis process itself, since cooling water, for example, can be returned to the environment via condensation. Table 4 provides a brief overview of the large-scale water demand.

Table 4: Water demand for electrolysis [31].

Water withdrawal from the environment:	
Alkaline electrolysis	32 Liter/kg _{H2}
PEM electrolysis	26 Liter/kg _{H2}
Water consumption:	
Alkaline electrolysis	22 Liter/kg _{H2}
PEM electrolysis	18 Liter/kg _{H2}

For decentralized applications in temperate climate zones, the use of drinking water is likely to pose fewer problems for two reasons. First, due to the small power capacity of such systems, the annual water demand is quite low. Second, decentralized usage implies that the hydrogen is produced and consumed on-site. Consequently, water is generated in the same area by consumption of hydrogen and thus remains within the local water cycle. Additionally, operating a decentralized water purification system for surface water or seawater would not be economically viable at this scale. Based on the values provided in Table 4, a water requirement of 20 L/kg_{H₂} is assumed for modeling purposes. Another ecological aspect is land use. Compared to maize-based biogas plants, photovoltaic (PV) ground-mounted systems require significantly less land area [32]. Furthermore, the operation of PV systems can be combined with agricultural use (agrivoltaics) [33] or ecological measures such as the creation of flower strips [34]. However, these aspects are not within the scope of the present study.

3.3 Example use case

The example use case is the application of hydrogen for fuel cell vehicles in agriculture. at the location of the city of Emden. The underlying data cover the period from 2010 to 2019 [1], ensuring that both seasonal variations and the impact of year-to-year differences are reflected in the results. In a first step, the required system size was determined. The demand for a 'small tractor' was set to 20 kg of hydrogen per working day (see Table 1). The simulation assumes three full working days per week for the tractor. This could theoretically correspond to use in a livestock operation and small-scale field work. The target hydrogen demand is assumed to be 20 kg, withdrawn from the storage tank on Mondays, Wednesdays, and Fridays. This scenario is based on the assumption that the tractor is primarily needed during the vegetation period (spring to autumn). For a rough estimation of the energy requirement, the average daily demand can be calculated as $3 * 20 \text{ kg} / 7 = 8.6 \text{ kg}$ of hydrogen per day. Assuming a minimum utilization of six months per year, the annual hydrogen demand amounts to 1 564 kg (equivalent to approx. 61 600 kWh or 6 160 liters of diesel fuel). In this use case, the consumption of hydrogen coincides with the period of highest photovoltaic production. In this scenario, the requirement is that the tractor should remain operational for as long a period as possible, while also maintaining economic viability. Table 5 summarizes the optimal sizes of the individual system components.

Table 5: Example scenarios.

Scenario	PV in kW	Wind in kW	Elektrolyzer in kW Compressor in kW Battery in kWh	Tank in kg (H ₂)
Tractor A)	200	–	50	80
Tractor B)	–	150	50	80
Tractor C)	100	100	50	100

Local wind power and photovoltaics

In general, wind power and photovoltaics differ significantly in their generation profiles. Photovoltaic systems are subject to both the day-night cycle and an annual cycle, with the maximum daylight duration reaching 17 hours at the summer solstice and dropping to a minimum of 7.5 hours at the winter solstice (for the city of Emden). In contrast, such strongly structured seasonal patterns are not typically observed for wind power. Although seasonal variations for wind do exist, they are less pronounced. Additionally, this model considers small-scale wind turbines, which, due to their lower tower height, are more exposed to turbulence near the ground. This represents a significant difference compared to large wind parks, where the much taller towers result in a far more stable and consistent generation profile.

Figure 7 shows the simulated annual electricity generation profile for a 40 kW photovoltaic (PV) system and a 40 kW wind turbine at the Emden site in 2012. For photovoltaics, a clear peak in summer is evident. In contrast, wind power generation shows smaller monthly fluctuations, with a distinct maximum in January – attributed to particularly stormy weather conditions in 2012.

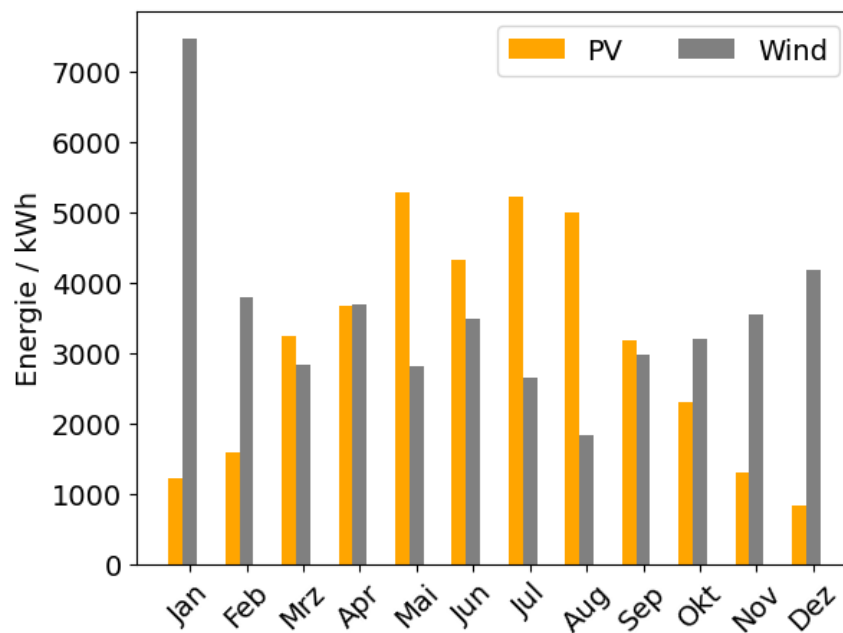


Figure 7: Wind power and photovoltaics production (each 40 kW) simulated for the year 2012.

On an annual average, both systems generate roughly the same amount of energy, as summarized in Table 6. These values are averaged over a 10-year period (2010–2019). At the same installed capacity, wind power produces about 10% more energy than photovoltaics. However, due to higher investment costs, decentralized wind power is approximately three times more expensive than decentralized solar power.

Table 6: Average annual production for the years 2010-2019.

	Annual production	Costs
Wind power 40 kW	44344 kWh	0.18 EUR/kWh
Photovoltaics 40 kW	38788 kWh	0.05 EUR/kWh

Electromobility

Here, the results for the application of hydrogen in agriculture (see Table 5) are presented. The high energy demand of tractors primarily occurs during the vegetation period, which coincides with the peak solar energy production. Table 7 provides an overview of the average annual energy production over the considered 10-year period. For the absolute amount of hydrogen produced, it does not matter whether the energy is supplied via photovoltaics or wind power – both systems can generate roughly the same total hydrogen output. However, the two energy sources have different availability patterns. Therefore, combining PV and wind power (case C) allows for a further increase in hydrogen production while simultaneously reducing grid feed-in.

Table 7: Average annual energy production for the years 2010-2019.

	Tractor A)	Tractor B)	Tractor C)
PV / kWh	193 710 (100%)	–	96 855 (47%)
Wind / kWh	–	163 127 (100%)	108 751 (53%)
Grid feed-in / kWh	59 840 (31%)	30 355 (19%)	29 380 (14%)
Hydrogen / kg	2 231	2 213	2 937

Figure 8 shows the temporal evolution of the storage level over the 10-year period under consideration. At first glance, the graph appears somewhat cluttered due to numerous fluctuations occurring over shorter time intervals. However, for case A) (PV-only), ten distinct maxima can be clearly identified, corresponding to the summer months. The maximum storage capacity is reached multiple times each year across all three cases A)–C). Case A) exhibits the highest variability. For the purely photovoltaic-based production, a significant drop in hydrogen production is observed in the years 2013 and 2016 due to local weather phenomena.

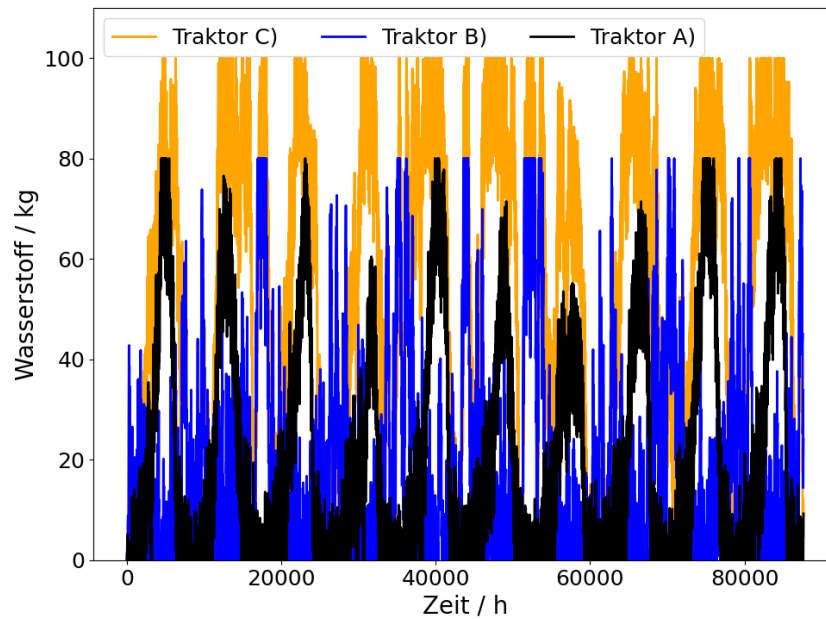


Figure 8: Level of hydrogen storage for the period 2010-2019; Capacity of storage A) = 80 kg B) = 80 kg C) = 100 kg

In order to assess the impact of a 'good' versus a 'bad' energy year, the availability of the tractor is evaluated. According to the model's assumption, the tractor should be refueled with 20 kg of hydrogen on Mondays, Wednesdays, and Fridays. In this context, availability indicates the percentage of these scheduled refueling days that can actually be fulfilled. This metric is equivalent to the vehicle's operational readiness. Figure 9 shows the availability for the year 2013, which was a 'poor' year for photovoltaic output. For case A) (shown in black, PV-only), the availability exceeds 80% from April to September. This means the tractor is not fully available for soil preparation and sowing in March, and similarly, it would not be reliably available for a potential harvest in October. For case B) (blue, wind power only), a 'summer gap' is observed from May to September, during which availability is significantly reduced. This period corresponds closely to the main vegetation phase and is therefore not acceptable for agricultural operations. In contrast, the combination of wind power and photovoltaics (orange, case C)) achieves a consistently high availability of over 80% throughout the year – except for January – and thus represents the best option.

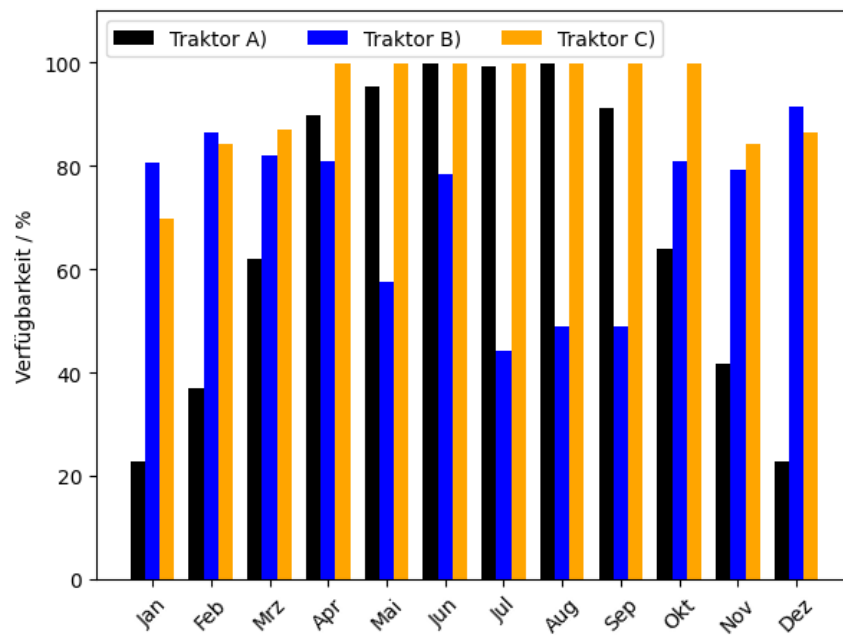


Figure 9: Driving range (i.e. availability) for the year 2013

Figure 10 shows the results for a 'good' year for photovoltaic production. Although a larger absolute amount of hydrogen can be produced in this case, the availability of the tractor across the months does not change significantly. Thus, one can conclude that the temporal availability of the tractor is quite independent of a 'good' or 'bad' energy year, provided that the system is properly dimensioned.

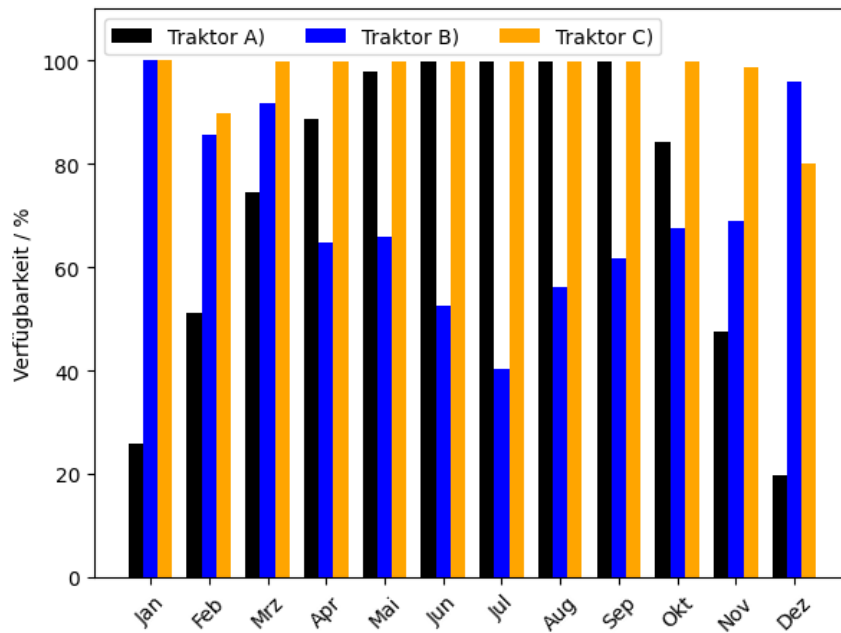


Figure 10: Driving range (i.e. availability) for the year 2018

Table 8 summarizes the costs associated with hydrogen use. The lowest costs are achieved with the combination of photovoltaics and wind power (case C)). This configuration enables the most efficient use of generated energy, which was already indicated by the low share of grid feed-in (see Table 7). For evaluating production costs, the price of diesel fuel serves as the most important reference point. In 2014, the price of subsidized agricultural diesel in Germany was approximately 1.50 EUR/L (0.15 EUR/kWh). Due to current global political developments, the price of diesel in German fuel stations reached around 2.00 EUR/L (0.20 EUR/kWh) in spring 2026. The future development of this price is subject to significant uncertainty. Currently, the production costs from Table 8 are roughly in the same range as those of diesel at the public fuel stations (for Germany).

Table 8: Average annual production costs for the years 2010-2019.

	Tractor A)	Tractor B)	Tractor C)
Production / EUR/kWh	0.22	0.23	0.17
Storage / EUR/kWh	0.02	0.02	0.02
Total cost / EUR/kWh	0.24	0.25	0.19
Production / EUR/kg	8.74	8.95	6.72
Storage / EUR/kg	0.73	0.74	0.69
Total costs / EUR/kg	9.47	9.68	7.42

The environmental balance of the three cases A)–C) is summarized in Table 9. Water consumption is a frequently discussed parameter. For comparison, the annual water usage of a single family home is about 100-200 m³. In this context, operating one or more small tractors does not impose any significant stress on local water management. The CO₂ reduction (based on crude oil) refers to the equivalent amount of diesel fuel that a conventional tractor would consume.

Table 9: Ecological analysis for the period of 2010-2019, regarding the usage of hydrogen.

	Tractor A)	Tractor B)	Tractor C)
Water demand / m ³	44.62	44.26	58.74
CO ₂ reduction (crude oil)/ tCO ₂	51.53	43.39	54.69
CO ₂ reduction (natural gas)/ tCO ₂	39.13	32.95	41.53

4 Appendix

4.1 Calculation of the energy demand for the compression of hydrogen

For a more accurate estimation, the compression work can be calculated as an ideal process of isothermal compression. This represents a thermodynamic limit – the absolute minimum amount of work required to compress the gas to the desired pressure. However, mechanical compressors, which are typically used in smaller-scale systems, usually perform an adiabatic process, meaning the gas heats up during compression and may require additional cooling afterward. The energy demand for isothermal compression W_i can be derived from the ideal gas law [15, 35]. The expression for isothermal compression is obtained accordingly as given in Equation 4.

$$W_i = R \cdot T \cdot \ln \frac{p_1}{p_2} \quad (4)$$

The adiabatic compression work W_a can be calculated according to Equation 5 [35],

$$W_a = k/(k-1) \cdot R \cdot T \cdot \left(\left(\frac{p_1}{p_2} \right)^{(k-1)/k} - 1 \right) \quad (5)$$

where $k = C_p/C_v$ is the heat capacity ratio. The following values were used for the equations above: $R(\text{H}_2) = 1.1548 \text{ Wh kg}^{-1} \text{ K}^{-1}$, $T = 293 \text{ K}$, $C_p = 14.199 \text{ kg K}$ und $C_v = 10.074 \text{ kg K}$.

Table 10 presents a comparison of the energy requirements for isothermal and adiabatic compression. The values obtained from Equations 4 and 5 show good agreement with literature data [36]. The actual energy demand for compression lies between the values for adiabatic and isothermal compression. While isothermal compression represents a theoretical minimum (a thermodynamic limit), it may be approximately approached by certain compressor designs [16, 36, 37, 38]. In this study, the adiabatic compression value is used as a conservative upper estimate. On one hand, the real energy demand for compression typically falls between the two values listed in Table 10 [36, 37]. On the other hand, the available literature does not provide specific efficiency data for the compressors under consideration.

Table 10: Energy demand of hydrogen compression.

p Tank / bar	p compressor / bar	Isothermal / kWh kg ⁻¹	Adiabatic / kWh kg ⁻¹
200.0	250.0	1.9	4.6
350.0	437.5	2.1	5.7

4.2 German energy efficiency ranking for residential buildings

Energy efficiency classes are a classification of buildings based on their annual heating energy demand. They are defined in the German Building Energy Act [22].

Table 11: German energy efficiency classes [22].

Efficiency class	Annual heat demand kWh m ⁻²
A+	≤ 30
A	≤ 50
B	≤ 75
C	≤ 100
D	≤ 130
E	≤ 160
F	≤ 200
G	≤ 250
H	> 250

4.3 Literature survey for CAPEX

Table 12 provides an overview of the investment costs for individual system components as available in the literature.

Table 12: Investment costs (CAPEX).

Component	Costs	Source
Photovoltaics	1500 EUR/kW	[24, 39]
	1380-1030 EUR/kW	[25] für 5-10 kWp
Wind power	1000 EUR/kW	[40, 41] offshore, >1 MW
	2700 EUR/kW	[40, 41, 42] onshore, >1 MW
	approx. 3000-5000 EUR/kW	[9] onshore, <500 kW
Elektrolyzer	1500 EUR/kW	[24]
	1500 EUR/kW	[43] PEM system
	1000-1500 EUR/kW	[44, 45] Study for development of prices
	1000 EUR/kW	[11] für 5 MW
	1000 EUR/kW	[43] alkaline system
Battery	800-625 EUR kWh	[46] for 40-75 kWh
	1000-500 EUR kWh	[47]
	800-600 EUR kWh	[48] for 5-12 kWh
	600-400 EUR kWh	[49] for 4-12 kWh
	200 EUR kWh	[24]
Compressor	5000 EUR/kW	[50] for 50 kW system
	1688 EUR/kW	[51] small fueling station at 250 bar
	1500-1000 EUR/kW	[50] for 100 kW system
	900 EUR/kW	[24] theoretical costs
	37 EUR/kW	[52] for 100 MW system
Tank	7-10 EUR/kg	[53] storage in salt caverns
	120 EUR/kg	[27] 350 bar for automotive (production costs)
	280 EUR/kg	[54] DOE (USA) cost target for mobile (2025)
	350 EUR/kg	[53] EU FCH-JU cost target for mobile (2024)
	115-600 EUR/kg	[54] mobile application
	450 EUR/kg	[55] steel bottle 50 l / 200 bar
	500 EUR/kg	[53] pressure storage (not underground)
	700 EUR/kg	[51] small fueling station 250 bar

References

- [1] European Commission and EU Science Hub. *PHOTOVOLTAIC GEOGRAPHICAL INFORMATION SYSTEM*. website accessed 30.01.2025. $\Rightarrow$ URL.
- [2] Python Software Foundation. *Python*. accessed 10.01.2025. $\Rightarrow$ URL.
- [3] The Matplotlib development team. *Matplotlib: Visualization with Python*. accessed 10.01.2025. $\Rightarrow$ URL.
- [4] Uwe Reimer. *MORE - Modelling Of Renewable Energy*. website accessed 20.07.2026. 2026. $\Rightarrow$ URL.
- [5] Lucas Bauer and Silvio Matysik. *Enercon E-18*. website accessed 18.12.2024. $\Rightarrow$ URL.
- [6] S. Karlsson. "The applicability of wind profile formulas to an urban-rural interface site". In: *Boundary-Layer Meteorology* 34 (1986), p. 333. $\Rightarrow$ URL.
- [7] E. Hau. *Windkraftanlagen - Grundlagen. Technik. Einsatz. Wirtschaftlichkeit*. 6th ed. Munich, Germany: Springer, 2016.
- [8] T. Uchida. "A New Proposal for Vertical Extrapolation of Offshore Wind Speed and an Assessment of Offshore Wind Energy Potential for the Hibikinada Area, Kitakyushu, Japan". In: *Energy and Power Engineering* 10 (2018), p. 154. $\Rightarrow$ URL.
- [9] U. Hallenga et al. *Kleinwindanlagen – Handbuch der Technik, Genehmigung und Wirtschaftlichkeit kleiner Windräder*. 2nd ed. Berlin: Bundesverband WindEnergie e.V., 2013. $\Rightarrow$ URL.
- [10] H. Wahl and A. Fübekker. *Viel Wind um nichts oder lohnende Investition - Kleinwindanlagen für den landwirtschaftlichen Betrieb?* Tech. rep. Webcode: 01043374. abgerufen am 30.03.2026. Landwirtschaftskammer Niedersachsen. $\Rightarrow$ URL.
- [11] M. Holst et al. *COST FORECAST FOR LOW-TEMPERATURE ELECTROLYSIS – TECHNOLOGY DRIVEN BOTTOM-UP PROGNOSIS FOR PEM AND ALKALINE WATER ELECTROLYSIS SYSTEMS*. Tech. rep. website accessed 21.04.2026. Fraunhofer Institute for Solar Energy Systems ISE, 2021. $\Rightarrow$ URL.
- [12] G. Squadrito, G. Maggio, and A. Nicita. "The green hydrogen revolution". In: *Renewable Energy* 216 (2023), p. 119041. $\Rightarrow$ URL.
- [13] R. Dufo-López, J. M. Lujano-Rojas, and J. L. Bernal-Agustín. "Optimisation of size and control strategy in utility scale green hydrogen production systems". In: *International Journal of Hydrogen Energy* 50 (2024), p. 292. $\Rightarrow$ URL.
- [14] D. Simbeck and E. Chang. *Hydrogen Supply: Cost Estimate for Hydrogen Pathways - Scoping Analysis*. Tech. rep. website accessed 21.04.2026. National Renewable Energy Laboratory (NREL), 2002. $\Rightarrow$ URL.
- [15] A. Züttel. "Hydrogen storage methods". In: *Naturwissenschaften* 91 (2004), p. 157. $\Rightarrow$ URL.
- [16] J. Jepsen. "Technical and Economic Evaluation of Hydrogen Storage Systems based on Light Metal Hydrides". HZG Report 2014-2; ISSN 2191-7833. Dissertation. Helmholtz-Zentrum Geesthacht, Zentrum für Material- und Küstenforschung GmbH: Helmut-Schmidt-Universität/Universität der Bundeswehr Hamburg, 2014.

-
- [17] H. Barthélémy. "Hydrogen storage - Industrial perspectives". In: *International Journal of Hydrogen Energy* 37 (2012), p. 1736. ⇒ URL.
- [18] Österreichisches Kuratorium für Landtechnik und Landentwicklung (OEKL). *Kraftstoffverbrauch in der Land- und Forstwirtschaft*. website accessed 30.01.2025. ⇒ URL.
- [19] H2 MOBILITY Deutschland GmbH und Co KG. *Hydrogen vehicles*. website accessed 30.01.2025. ⇒ URL.
- [20] P. Reithuber et al. "Energy and hydrogen consumption evaluation of a fuel cell city bus based on roller chassis dynamometer measurements". In: *International Journal of Hydrogen Energy* 97 (2025), p. 1227. ⇒ URL.
- [21] ARTHUR BUS GmbH. *ARTHUR BUS*. website accessed 17.04.2026. ⇒ URL.
- [22] Bundesministerium der Justiz und für Verbraucherschutz. *Gesetz zur Einsparung von Energie und zur Nutzung erneuerbarer Energien zur Wärme- und Kälteerzeugung in Gebäuden*. website accessed 21.04.2026. 2020. ⇒ URL.
- [23] S. Becker et al. *DENA Gebäudereport 2024: Zahlen, Daten, Fakten zum Klimaschutz im Gebäudebestand*. Tech. rep. website accessed 21.04.2026. Deutsche Energie-Agentur GmbH (DENA), 2023. ⇒ URL.
- [24] G. Erichsen et al. *Data Documentation: VEREKON Cost Parameters for 2050 in its Energy System Model*. website accessed 30.09.2024. 2019. ⇒ URL.
- [25] Musterhaus.net GmbH. *Kosten und Preise für Photovoltaik im Jahr 2024*. website accessed 07.10.2024. 2024. ⇒ URL.
- [26] IEA. *Projected Costs of Generating Electricity 2020*). Tech. rep. NEA No. 7531. website accessed 14.04.2026. International Energy Agency, 2020. ⇒ URL.
- [27] A. Kampker et al. *Economic and Ecological Analysis of Hydrogen Storage Systems*. Tech. rep. ISSN: 2701-6277. abgerufen am 20.04.2026. CONFERENCE ON PRODUCTION SYSTEMS and LOGISTICS - CPSL 2024. ⇒ URL.
- [28] Kristina Juhrich. *CO₂-Emissionsfaktoren für fossile Brennstoffe*. Tech. rep. website accessed 21.04.2026. Umweltbundesamt, 2016. ⇒ URL.
- [29] F. Altgelt et al. *Water Consumption of Powerfuels*. Tech. rep. website accessed 21.04.2026. Deutsche Energie-Agentur GmbH (DENA), 2021. ⇒ URL.
- [30] Emily Grubert. "Water consumption from electrolytic hydrogen in a carbon-neutral US energy system". In: *Cleaner Production Letters* 4 (2023), p. 100037. ⇒ URL.
- [31] Bluerisk. *Water for hydrogen production*. Tech. rep. website accessed 21.04.2026. International Renewable Energy Agency (IRENA), 2023. ⇒ URL.
- [32] U. Würflein et al. *Photovoltaik auf dem Acker? - Ein Positionspapier*. Tech. rep. Umweltbundesamt, 2025. ⇒ URL.
- [33] C. J. Martínez-Hernández et al. "Efficiency, Sustainability and Governance of Agrivoltaic Systems: A PRISMA-Based Systematic Review of Global Evidence (2010–2025)". In: *Energies* 19 (2026), p. 1418. ⇒ URL.

-
- [34] K. Wunder et al. "Environmentally friendly solar energy: potential of utilizing field margins in rural areas for photovoltaic systems in combination with wildflower strips". In: *22nd Wind and Solar Integration Workshop (WIW 2023)*. Vol. 2023. 2023, pp. 775–782. ⇒ URL.
- [35] G. Wedler. *Lehrbuch der Physikalischen Chemie*. 3rd ed. Weinheim: VCH, 1987.
- [36] Sofoklis S. Makridis. "Hydrogen storage and compression". In: *Methane and hydrogen for energy storage*. Ed. by R. Carriveau and David S.-K. Ting. Windsor, Ontario, Canada: Institution of Engineering and Technology, 2016, pp. 1–28. ⇒ URL.
- [37] M.-R. Tahan. "Recent advances in hydrogen compressors for use in large-scale renewable energy integration". In: *International Journal of Hydrogen Energy* 47 (2022), p. 35275. ⇒ URL.
- [38] A World Of Energy. *Hydrogen compression*. website accessed 10.10.2024. 2024. ⇒ URL.
- [39] Wegatech Greenenergy GmbH. *So viel kostet eine Photovoltaikanlage*. website accessed 07.10.2024. 2024. ⇒ URL.
- [40] R. Satymov, D. Bogdanov, and C. Breyer. "Techno-economics of offshore wind power in global resolution". In: *Applied Energy* 393 (2025), p. 125980. ⇒ URL.
- [41] International Renewable Energy Agency (IRENA). *Renewable power generation costs in 2023*. website accessed 14.04.2026. ⇒ URL.
- [42] S. Lüers and A.-K. Wallasch. *Kostensituation der Windenergie an Land Stand 2023 (Auszug aus dem Gesamtbericht vom 22. Juni 2023)*. Tech. rep. Berichtsnr.: SP23010A1 / FKZ: 03MAP393. Deutsche WindGuard GmbH im Auftrag des Bundesministeriums für Wirtschaft und Klimaschutz, 2023. ⇒ URL.
- [43] Joris Proost. "State-of-the art CAPEX data for water electrolyzers, and their impact on renewable hydrogen price settings". In: *International Journal of Hydrogen Energy* 4 (2019), p. 4406. ⇒ URL.
- [44] Andreas Luczak. *Zukünftige Erzeugung und Nutzung von grünem Wasserstoff in Schleswig-Holstein: Abschätzung der CO₂-Vermeidungskosten*. Tech. rep. website accessed 21.04.2026. Gesellschaft für Energie und Klimaschutz Schleswig-Holstein GmbH, 2022. ⇒ URL.
- [45] G. Glenk and S. Reichelstein. "Economics of converting renewable power to hydrogen". In: *Nature Energy* 4 (2019), p. 216. ⇒ URL.
- [46] LadeEngel Inh. Oliver Kalz e.K. *Batteriespeicher bis 75 kWh*. website accessed 23.04.2026. 2026. ⇒ URL.
- [47] Verbraucherzentrale NRW e.V. *Lohnen sich Batteriespeicher für Photovoltaik-Anlagen?* website accessed 07.10.2024. 2024. ⇒ URL.
- [48] Wegatech Greenenergy GmbH. *Kosten und Preise von Stromspeichern*. website accessed 07.10.2024. 2024. ⇒ URL.
- [49] Musterhaus.net GmbH. *Stromspeicher-Kosten: Preise für PV-Speicher in 2024*. website accessed 07.10.2024. 2024. ⇒ URL.
- [50] S. Niaz, T. Manzoor, and A. H. Pandith. "Hydrogen storage: Materials, Methods and Perspectives". In: *Renewable and Sustainable Energy Reviews* 50 (2015), p. 457. ⇒ URL.

-
- [51] Ø. Ulleberg and R. Hancke. "Techno-economic calculations of small-scale hydrogen supply systems for zero emission transport in Norway". In: *International Journal of Hydrogen Energy* 45 (2020), p. 1201. ⇒ URL.
- [52] K. Mongird et al. *2020 Grid Energy Storage Technology Cost and Performance Assessment*. Tech. rep. DOE/PA-0204. Pacific Northwest National Laboratory, 2020.
- [53] C. Moran et al. "The hydrogen storage challenge: Does storage method and size affect the cost and operational flexibility of hydrogen supply chains?" In: *International Journal of Hydrogen Energy* 52 (2024), p. 1090. ⇒ URL.
- [54] H. K. Shin and S. K. Ha. "A Review on the Cost Analysis of Hydrogen Gas Storage Tanks for Fuel Cell Vehicles". In: *Energies* 16 (2023), p. 5233. ⇒ URL.
- [55] Gase Dopp / Fa. Frank Dopp. *Wasserstoff - 50 Liter Flasche 200 bar neu + ungefüllt*. website accessed 23.04.2026. 2026. ⇒ URL.